

Implementation and Evaluation of Near-Field Integrated Sensing, Computing and Semantic Communication for Digital Twin-Assisted Vehicular Networks

CS741: Next Generation Cloud Architecture

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CERTIFICATE

This is to certify that the research assignment report entitled **“Implementation and Evaluation of Near-Field Integrated Sensing, Computing and Semantic Communication for Digital Twin-Assisted Vehicular Networks”** is a bona fide record of the work carried out by Yagnang Acharya (262IS037), Akash Minz (262IS002), Amul Gourav (262IS005), in partial fulfilment of the requirements of the course Next Generation Cloud Architecture (CS741), Department of Computer Science and Engineering, National Institute of Technology Karnataka, Surathkal, during the academic session 2026-27, under my guidance.

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DECLARATION

We hereby declare that the work presented in this report is our own. The source code was written by us from the equations of the reference paper and has not been copied from any existing implementation. Wherever we have used the work of others, it has been cited. All results reported here were generated by our own code, and the raw numbers are available in the project repository.

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ABSTRACT

Roadside units (RSUs) with very large millimetre-wave antenna arrays can sense the vehicles on a road with the same signal they use to communicate with them. Yang et al. (IEEE Transactions on Vehicular Technology, 2026) propose a design that integrates sensing, computing for a digital twin (DT), and semantic communication (ISCSC) in the near field of such arrays, and claim about 20% higher transmission rate than conventional integrated sensing and communication (ISAC) at equal sensing accuracy.

In this assignment we reproduced the complete system in Python directly from the paper's equations and parameter table. We implemented the near-field channel, a highway scenario with randomly placed vehicles, three tracking filters (particle filter with 500 and 2000 particles, extended and unscented Kalman filters), the Cramér-Rao bound (CRB), vehicle-to-RSU assignment by simulated annealing, joint beamforming and semantic-ratio optimisation, and the conventional ISAC benchmark of the paper's reference [18]. The work was carried out in eight staged steps, each verified by automated tests (189 tests, all passing) and run over five random seeds.

The near-field region of about 300 m is confirmed (288 m), the particle filter's velocity error of 0.8 m/s is reproduced, the CPU-versus-delay behaviour of the paper's Fig. 8 is reproduced, and the proposed hybrid heuristic outperforms greedy assignment as claimed. Several absolute values are not reproduced: achieved rates are four to five times higher, distance accuracy behaves differently, and the EKF and UKF track as well as the particle filter. The headline claim is **not reproduced** as stated: for 1 Kb tasks the gain is 36% rather than 20%, and it arises entirely from the semantic factor $\iota/\rho = 1.1/0.81$ in the rate expression, not from improved beamforming or assignment. For 10 Kb tasks the system is feasible only for five vehicles, and for 100 Kb tasks it is infeasible under the paper's own CPU and latency limits.

Keywords: near-field communication, integrated sensing and communication, semantic communication, digital twin, vehicular networks, particle filter, Cramér-Rao bound, simulated annealing, reproducibility.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AO	Alternating optimisation
CPU	Central processing unit
CRB / RCRB	Cramér-Rao bound / root Cramér-Rao bound
DT	Digital twin
EKF	Extended Kalman filter
FIM	Fisher information matrix
HH	Hybrid heuristic (greedy + simulated annealing, Algorithm 2)
ISAC	Integrated sensing and communication
ISCSC	Integrated sensing, computing and semantic communication
L-BFGS	Limited-memory Broyden-Fletcher-Goldfarb-Shanno (optimiser)
MIMO	Multiple-input multiple-output
ML	Maximum likelihood
mmWave	Millimetre wave
PF	Particle filter
RMSE	Root mean square error
RSU	Roadside unit
SA	Simulated annealing
SDR	Semidefinite relaxation
SNR	Signal-to-noise ratio
UKF	Unscented Kalman filter
ULA	Uniform linear array

CHAPTER 1

Introduction

1.1 Background

A **digital twin** of a road is a live software replica of every vehicle on it, holding each vehicle's position and velocity. Intelligent transport systems use such a twin to plan traffic, issue warnings and support driving decisions. The twin must be kept up to date, which needs both accurate sensing of the vehicles and enough computing to process the data within a strict time limit.

At millimetre-wave frequencies, roadside units can carry very large antenna arrays. Such an array can do two jobs with one signal: it can communicate with the vehicles, and the echoes of the same signal reveal where the vehicles are. This is called **integrated sensing and communication (ISAC)**. When the array is large and the vehicles are close, the vehicles lie in the array's **near field**, where the wavefront is spherical rather than flat. Modelling this curvature makes beamforming harder, but it lets a single RSU estimate both the angle and the distance of each vehicle.

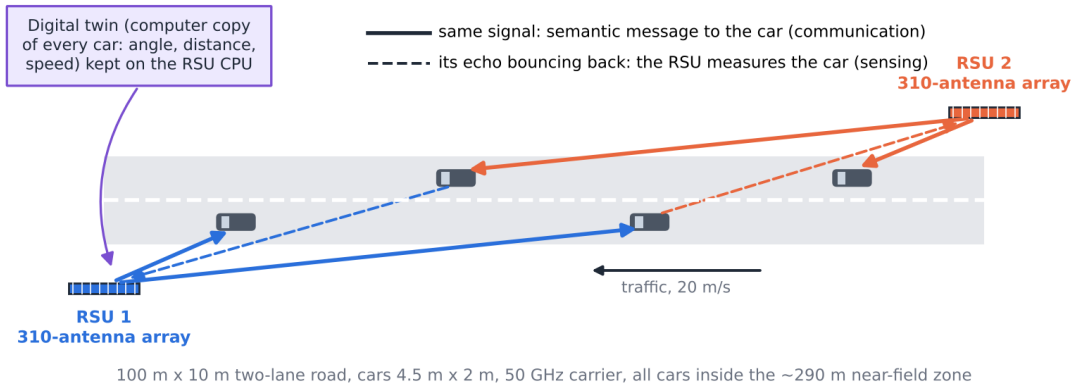


Figure 1.1: Overview of the system. A large antenna array beside the road communicates with the vehicles and, with the same signal, senses them.

1.2 Motivation

In Task 1 of this course the digital twin was filled with data that each vehicle senses itself and uploads. Task 2 studies the alternative in which the RSU senses the vehicles directly. If the RSU can sense and communicate with one signal, the vehicles need not upload their own sensing data, and the same spectrum serves both purposes. The reference paper adds **semantic communication**, in which a neural encoder sends the meaning of a message rather than every bit, so that fewer bits are needed on air at the cost of extra computing power. Whether this combination really gives the claimed rate gain is the central question of this assignment.

1.3 Problem Statement

The task is to reproduce the integrated sensing, computing and semantic communication (ISCSC) framework of Yang et al. [1] for digital twin-assisted vehicular networks. The framework must solve three coupled problems at once: (i) track the state of each vehicle accurately enough to maintain the twin; (ii) allocate the RSU's computing resources across vehicles; and (iii) design the beamformers and semantic extraction ratios so that both sensing accuracy and communication rate requirements are met. The reproduction must state clearly which of the paper's results are reproduced and which are not, and evaluate the cost of the approach relative to Task 1.

1.4 Objectives

- Reproduce the near-field system model: RSUs with a uniform linear array of Nt half-wavelength-spaced antennas in a MU-MIMO mmWave configuration, with near-field coverage of about 300 m; vehicles placed by a Poisson process on a 100 m × 10 m highway, vehicle size 4.5 m × 2 m, minimum spacing 3.5 m.
- Implement the near-field steering vector and channel model, compare it with the far-field approximation and identify the range at which they diverge.
- Implement and compare the particle filter (Ns = 500 and 2000), EKF and UKF, measuring angle, distance and velocity RMSE and computation time as the number of vehicles grows.
- Implement the resource allocation and beamforming optimisation, including simulated annealing for vehicle-to-RSU assignment, and evaluate the angle CRB as the sensing benchmark.
- Test the headline claim of a 20% rate gain over conventional ISAC at equal sensing accuracy, at data sizes of 1 Kb, 10 Kb and 100 Kb.
- Deliver a documented Python implementation, a reproduction report and the required measurements.

1.5 Scope of the Work

The work covers the complete signal-level and optimisation-level model of the paper, simulated in Python. The semantic encoder is represented, as in the paper, only through the extraction ratio ρ and its power cost; no neural network is trained. Vehicle motion follows the paper's own motion model rather than an external traffic simulator. Values of the paper's figures were read by eye for comparison.

1.6 Organisation of the Report

Chapter 2 reviews the background techniques. Chapter 3 presents the system model and the optimisation problem as given in the paper. Chapter 4 describes our methodology, tools, software design and assumptions. Chapter 5 describes the implementation of each module.

Chapter 6 presents and discusses the results against the paper. Chapter 7 concludes the report with limitations and future scope. Appendices give the run instructions and the full measurement tables.

CHAPTER 2

Literature Review

2.1 Near-Field Communication and Sensing

The boundary between the near field and the far field of an antenna array is usually taken as the Rayleigh distance $2D^2/\lambda$, where D is the array aperture and λ the wavelength [5]. With hundreds of antennas at millimetre-wave frequencies this distance reaches hundreds of metres, so users and targets near an RSU are in the near field. The spherical wavefront makes the array response depend on both angle and distance, which allows beams to be focused on a point rather than a direction [5]. Wang, Mu and Liu [2] used this property to design near-field ISAC beamforming that minimises the CRB of a target's angle and distance under minimum-rate constraints for communication users; the reference paper uses this design as its benchmark.

2.2 Semantic Communication

Semantic communication transmits the meaning of a message instead of its exact bit sequence. In the reference paper this is captured by a semantic rate equal to (ι/ρ) times the Shannon rate, where ι converts words to bits and ρ is the semantic extraction ratio; stronger compression (smaller ρ) increases the effective rate but costs more computing power [1].

2.3 Vehicle Tracking Filters

The particle filter represents the distribution of a vehicle's state by weighted samples that are predicted with the motion model, weighted by the measurement likelihood and resampled [6]. The extended Kalman filter linearises the motion and measurement models around the current estimate, while the unscented Kalman filter propagates a small set of deterministically chosen sigma points through the non-linear models [7]. The particle filter handles strong non-linearity best but costs more computation.

2.4 Cramér-Rao Bound

The CRB is the inverse of the Fisher information matrix and gives the lowest variance any unbiased estimator can achieve. It is widely used as the sensing metric in ISAC design because it can be written in closed form in terms of the transmit covariance [2]. For filters that use a motion model over many time steps, the appropriate bound is the posterior CRB of Tichavský et al. [3].

2.5 Optimisation Techniques

Beamforming problems with rank-one beam vectors are non-convex. Semidefinite relaxation (SDR) replaces the beam vectors by positive semidefinite matrices and drops the rank constraint, after which Gaussian randomisation recovers rank-one beams [8]. Large semidefinite problems can alternatively be solved through a low-rank factorisation [9], which we use in our solver. Assignment of vehicles to RSUs is combinatorial; simulated annealing [10] searches it by accepting worse moves with a probability that decreases over time, to avoid local optima.

2.6 The Reference Paper and the Research Gap

Yang et al. [1] combine all of the above into one ISCSC framework with a digital twin at the RSU. They report tracking results (Fig. 3), rate and CRB against the number of vehicles (Figs. 5-7), and computing frequency and power against data size (Figs. 8-9). No code is published with the paper, and several parameters and units are not stated. An independent reproduction is therefore needed to check whether the reported numbers, and in particular the 20% rate gain, follow from the stated model.

CHAPTER 3

System Model and Problem Formulation

This chapter summarises the model exactly as given in the reference paper [1]. Equation numbers refer to the paper.

3.1 Network Model

Two RSUs stand on either side of a straight two-lane highway of 100 m × 10 m. Each RSU has a 310-element half-wavelength uniform linear array at 50 GHz; each vehicle has 3 receive antennas. Time is divided into slots of 20 ms. In each slot every RSU transmits one joint signal containing communication beams for the vehicles it serves and sensing beams. Each vehicle is served by exactly one RSU but receives interference from all beams of both RSUs.

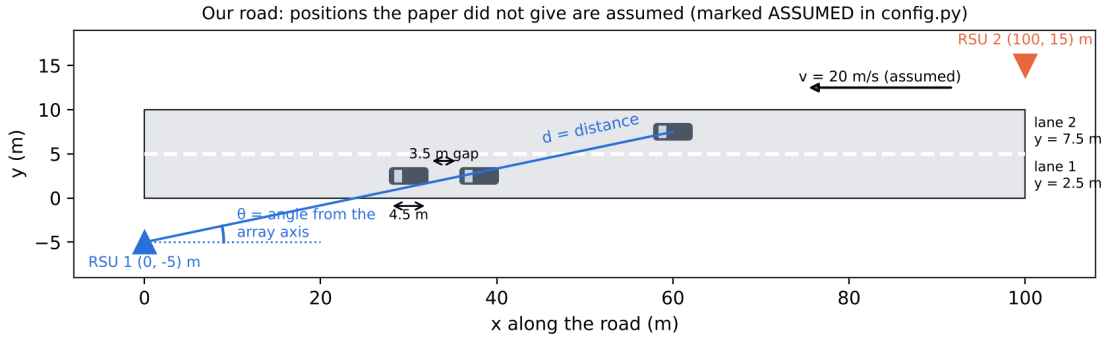


Figure 3.1: Geometry used in this work. The angle θ is measured from the array axis (the road direction) and the distance d from the array centre.

3.2 Near-Field Channel Model

The transmit and receive steering vectors follow the second-order Fresnel approximation of the spherical wavefront, so the phase across the array depends on both angle θ and distance d (Eq. 5). The per-element path loss is given by Γ (Eq. 6), and the received signal at a vehicle is the sum of the desired signal, interference from other vehicles' communication beams, interference from sensing beams, and noise (Eq. 4).

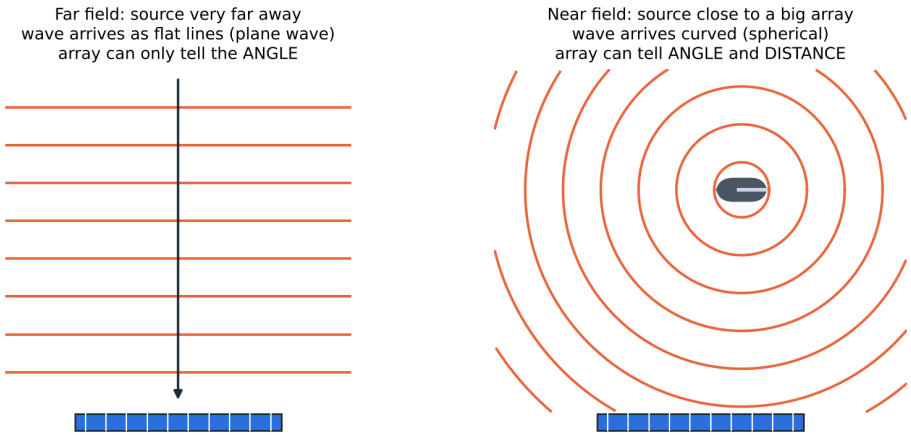


Figure 3.2: Far-field (plane) versus near-field (spherical) wavefronts. The curvature carries distance information.

3.3 Sensing and Tracking Model

The state of each vehicle is q = [θ, d, v, β], where β is the round-trip path loss. The motion model (Eq. 10) assumes nearly constant velocity along the road; the measurement model (Eq. 9) gives a noisy echo vector, noisy distance and noisy velocity. Process and measurement noises have covariances Q1 and Q2 (Eqs. 11-12). The paper tracks each vehicle with a particle filter (Algorithm 1, Eqs. 26-33).

3.4 Sensing Accuracy: the Cramér-Rao Bound

The Fisher information matrix for (d, θ, β) is given by Eqs. (15)-(18) in terms of the transmit covariance $Rx = WW^H + RR^H$ (Eq. 3). After removing β with the Schur complement, CRB(d) and CRB(θ) are the diagonal entries of the inverse (Eq. 19).

3.5 Semantic Rate

The semantic rate of vehicle k (Eq. 13) is (ι/ρ) multiplied by the log-determinant capacity of its MIMO link, with ι = 1.1 and a lower bound on ρ set by the required semantic similarity (Eq. 14).

3.6 Computing and Power Model

Semantic extraction costs power $P_{Comp} = -F\,\Sigma \ln \rho$ (Eq. 20); the beams cost $P_{C\&S} = \mathrm{Tr}(WW^H + RR^H)$ (Eq. 21). Updating the twin for vehicle k needs C·D + L CPU cycles within T^{max} (Eqs. 22-23), where the extra workload L grows with the root CRB of angle and distance (Eq. 24). The twin's CPU power is P_{DT} (Eq. 25). The total of the three must not exceed the RSU budget Pt.

3.7 Optimisation Problem

The paper's problem (34) chooses the vehicle-to-RSU assignment ξ, the beamformers W and R, the semantic ratios ρ and the CPU frequencies f to minimise a weighted sum of the negative semantic rate and the sum of CRBs (weight $\varepsilon = 0.5$), subject to the power budget (34e), CPU limit F^{max}, latency limit T^{max}, a minimum ρ, and one RSU per vehicle.

3.8 Solution Approach of the Paper

The paper splits (34) into two nested loops. The outer loop (Algorithm 2) starts from a greedy nearest-RSU assignment and improves it by simulated annealing with a tabu list. For each assignment, the inner loop (Algorithm 3) applies alternating optimisation: with ρ fixed, the beams are found from a convex problem (41) obtained by SDR, a log-determinant identity (Lemma 1), a low-SNR approximation (Lemma 2) and a Schur-complement form of the CRB constraint (40); then ρ is updated; the two steps repeat until the objective changes by less than 0.1. Gaussian randomisation finally gives rank-one beams.

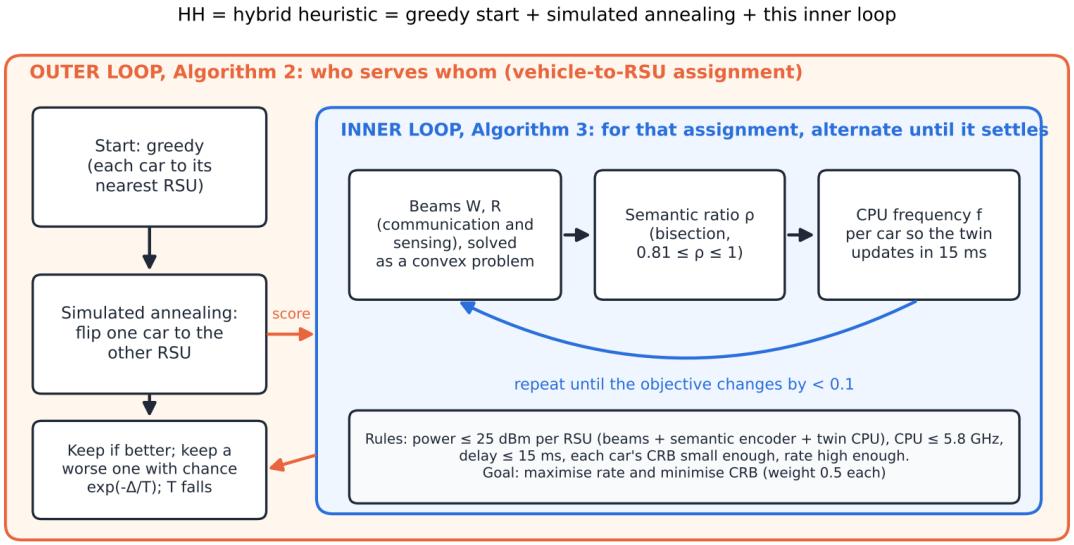


Figure 3.3: The paper's two nested loops: simulated annealing over assignments (outer) and alternating optimisation of beams, ρ and CPU (inner).

Table 3.1: Simulation parameters from the reference paper.

Symbol	Meaning	Value	Source
fc	Carrier frequency	50 GHz	Table III
Nt / Nr	RSU antennas / vehicle antennas	310 / 3	Table III
M / K	Number of RSUs / vehicles	2 / 5-20	Table III, Figs. 3, 5-7
ΔT	Slot length	20 ms	Table III
Pt	RSU power budget	25 dBm (316 mW)	Table III
σ_c^2, σ_r^2	Communication and echo noise	−30 dBm	Table III
ι / ρ	Word-to-bit factor / semantic ratio	1.1 / 0.81	Table III
F	Computing power coefficient	10 (unit not given)	Table III
Q1	Process noise (θ, d, v, β)	[0.02, 0.3, 1, 0.1] (units not given)	Table III
Q2	Measurement noise	[0.04, 0.06, 1] (units not given)	Table III
C / D	CPU cycles per bit / task size	1000-2000 / 1000-3000 bits	Table III
T^{max} / F^{max}	Latency limit / RSU CPU limit	15 ms / 5.8 GHz	Table III
κ	CPU energy coefficient	10^{-28}	Table III
ν1 / ν2 / ν3	Workload per unit RCRB / extra term	243.2 MHz/deg, 121.6 MHz/m, CN(0, 1000)	Table III
ε	Objective weight	0.5	Table III
T_0	Initial SA temperature	100	Algorithm 2
Ns	Particles	500, 2000	Sec. VI-A

CHAPTER 4

Methodology and System Design

4.1 Reproduction Principles

- Each model is implemented exactly as written in the paper, with the equation number quoted in the code.
- Every numeric parameter carries its source in `config.py`. Values the paper does not give are marked **ASSUMED** with a reason.
- No result, parameter or equation is invented. A claim is counted as reproduced only if our numbers show it.
- Every random process takes a seed; every experiment runs at least five seeds and reports mean and standard deviation; raw numbers are saved as CSV files.
- Every model has an automated test against an analytic value, a limiting case or a symmetry.
- SI units are used inside the code; degrees and dB appear only in plots and tables.

4.2 Staged Workflow

The work was divided into eight stages. A stage was started only after the previous stage's tests passed and its results were reviewed, and each stage was committed to the Git repository separately. This order was chosen because each stage depends on the one before it, and because it localises errors: for example, a sign error in the paper's Lemma 1 was caught by a test in Stage 5 instead of appearing as an unexplained rate in Stage 6.

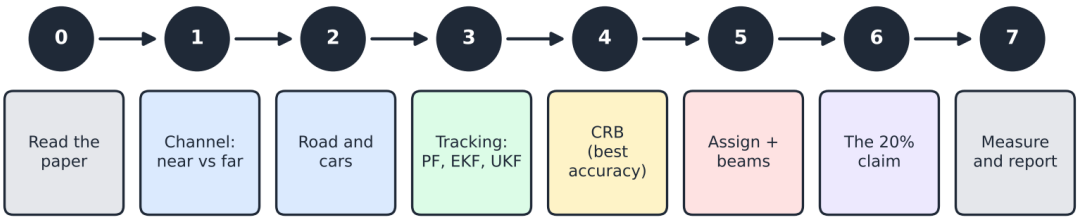


Figure 4.1: The eight stages of the work.

Table 4.1: Stages of the work and the objectives they address.

Stage	Work done	Objective
0	Read the paper; parameter table and list of gaps; no code	All
1	Near-field channel and comparison with far field	1, 2
2	Road, RSUs and vehicle placement	1
3	Tracking with PF (500, 2000), EKF and UKF	3
4	Cramér-Rao bound	4
5	Assignment, beamforming and computing	4
6	Headline claim at 1, 10 and 100 Kb	5
7	Measurements and report	6

4.3 Tools and Technologies

Table 4.2: Software tools used. Exact versions are pinned in requirements.txt.

Tool	Version	Purpose
Python	3.11	Implementation language
NumPy	2.2.6	Linear algebra, complex arithmetic, seeded random numbers
SciPy	1.15.3	L-BFGS optimiser, linear algebra, statistics
Pandas	2.2.3	Result tables, CSV output, mean and standard deviation
Matplotlib	3.10.9	Plots
pytest	8.4.2	Automated tests
CVXPY, SCS, CLARABEL	1.6.7, 3.2.11, 0.10.0	Cross-checking our beamforming solver on small cases only
Git, GitHub	-	Version control, one commit per stage

PyTorch was not required, since the semantic encoder enters the model only through ρ and Eq. (20). A traffic simulator such as SUMO was not required, since the paper's motion model (Eq. 10) is implemented directly.

4.4 Software Architecture

Figure 4.2 shows how the modules depend on each other. Parameters flow from `config.py` into the scenario, which gives the angle and distance of every vehicle from every RSU; the channel module builds the near-field channels; tracking, CRB and beamforming use these channels; and the experiment scripts produce every figure and table.

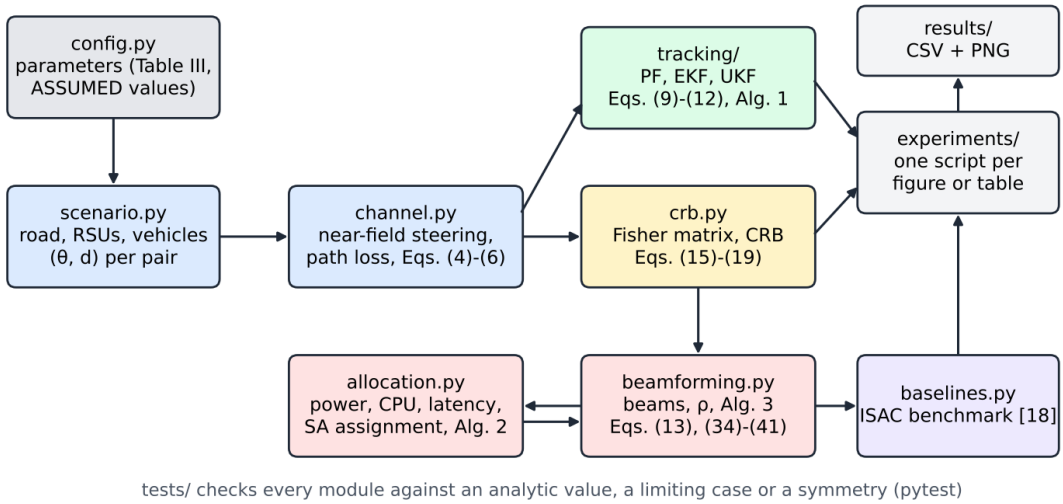


Figure 4.2: Software architecture and data flow between modules.

Table 4.3: Modules of the implementation and the equations they implement.

Module	Contents	Paper
`config.py`	All parameters with their sources	Table III
`channel.py`	Steering vectors, cross term, path loss, channel, received signal	Eqs. (4)-(6)
`scenario.py`	Road, lanes, RSU positions, vehicle placement	Sec. II, VI
`tracking/`	Motion and measurement models; PF, EKF, UKF	Eqs. (9)-(12), (26)-(33), Alg. 1
`crb.py`	Fisher matrix, CRB, posterior CRB	Eqs. (15)-(19)
`allocation.py`	Power, latency, DT workload, greedy and SA assignment	Eqs. (20)-(25), Alg. 2
`beamforming.py`	Rate, problem (41), alternating optimisation, randomisation	Eqs. (13), (34)-(41), Alg. 3
`baselines.py`	ISAC benchmark of [2]	[2] problem (16), (20), (21)
`experiments/`	One script per figure or table	Figs. 3, 5-9
`tests/`	189 automated tests	-

4.5 Assumptions

The paper leaves 36 details unstated; all are recorded with reasons in `docs/06-assumptions.md`. Those that affect the results most are listed in Table 4.4.

Table 4.4: Main assumptions and corrections.

Item	What the paper leaves open	Our choice and reason
RSU positions	Not given	RSU 1 at (0, −5) m, RSU 2 at (100, 15) m, arrays along the road; matches Fig. 2
Vehicle speed	Not given	20 m/s, same direction for all vehicles
Γ in Eqs. (4), (13)	Matrix sizes do not match	Applied element by element, the only consistent reading
Q1, Q2 units	Not given	Variances, angle entries in degree²; rad² gives 60° errors, far from Fig. 3
β noise	Variance 0.1, 100-300 times β itself	Relative noise, 32% of β per slot; as-written run kept
Eqs. (16), (17)	Real part missing; β* outside Re{·}	Corrected; a Fisher matrix must be real
σ_z^2, T in CRB	Not given	−30 dBm, T = 1
ρ = 0.81	Role unclear	Lower bound on ρ, starting value ρ = 1
F, Eq. (25)	Units make (34e) impossible	F in mW; $P_{DT} = \kappa f^2 CD/\Delta T$ (matches the shape of Fig. 9)
Latency, Eq. (24)	Never satisfiable with Table III	Main runs use L = |v3|; the as-written check is recorded
Lemma 1	Sign error	−ln|E| minimised directly
SA settings	Only T_0 given	Cooling 0.95, $T_{min} = 10^{-3}$, 200 iterations
Greedy with flip	Number of flips not given	Each vehicle switches RSU with probability 0.2
HH non-semantic	Not defined	ρ = 1, ι = 1, no extraction power
Benchmark	Algorithm not described	Implemented from [2], extended to two RSUs

4.6 Verification Strategy

Each module is tested independently before it is used by the next stage. Tests compare the code with analytic results (for example, the near-field vector must tend to the far-field vector at large distance), with brute-force alternatives (rejection sampling, finite-difference gradients, a general-purpose convex solver), and with statistical checks (a maximum-likelihood estimator must reach the CRB). Details are given in Section 5.7.

CHAPTER 5

Implementation

5.1 Near-Field Channel

The steering vectors of Eq. (5), the per-element path loss of Eq. (6) and the received signal of Eq. (4) are implemented in `channel.py`. For comparison we also implemented the exact spherical-wave vector and the plane-wave vector; the paper has no such comparison.

5.2 Road Scenario

For a fixed number of vehicles K, positions are drawn uniformly, since a Poisson process conditioned on its count is uniform. Plain rejection sampling is impractical: a uniform layout satisfies the 3.5 m spacing rule with probability 0.41 at K = 5 but only 1.5×10^{-13} at K = 20. We therefore implemented an exact one-dimensional hard-core sampler with the same distribution, verified against rejection sampling at K = 6.

5.3 Tracking Filters

The particle filter follows Algorithm 1 with multinomial resampling every slot. The EKF and UKF are the standard forms (the paper only cites them), using the same models; the UKF uses $\alpha = 10^{-3}$, $\beta = 2$, $\kappa = 0$. Each run tracks K vehicles from both RSUs for 100 slots (2 s). The true tracks are generated from Eq. (10) itself, since the first-order model drifts up to 0.48 m and 2.2° from an exact straight-line track over 2 s.

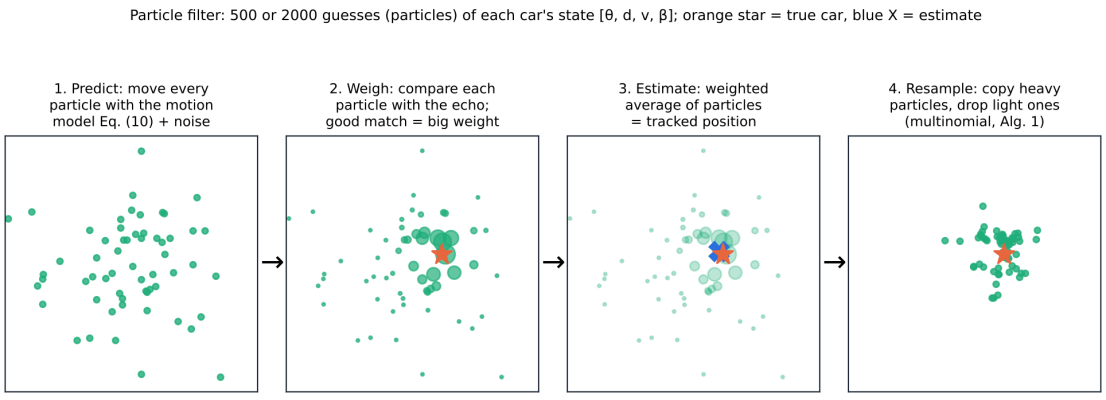


Figure 5.1: One step of the particle filter, repeated every 20 ms slot.

5.4 Cramér-Rao Bound

The CRB of Eqs. (15)-(19) was derived step by step from the Slepian-Bangs formula (`docs/stage4-notes.md`). Two corrections were needed for a valid Fisher matrix: Re{·} is missing in Eq. (16), and β* must lie inside Re{·} in Eq. (17). For checking the filters we also

implemented the posterior CRB [3].

5.5 Assignment and Beamforming

- **Assignment** follows Algorithm 2: greedy start, then simulated annealing with one-vehicle moves, using the negative of (41a) as the score so that the direction of search agrees with the minimisation.
- **Lemma 1** as printed gives $\ln|E|$ rather than $\ln|E^{-1}|$ (shown by a test); we minimise $-\ln|E|$ directly.
- **Solver.** CVXPY with SCS and CLARABEL is not accurate at the real SNR (SCS returned beams 31% over the power limit). We therefore solve (41) with L-BFGS on a low-rank factorisation $X = (P/S)YY^H$, which keeps the power and positive semidefiniteness exact, and we verified it against CLARABEL on low-SNR cases.
- **Dimension reduction.** The optimum lies in the span of the steering vectors and their derivatives, so each problem is reduced exactly to about 20-50 dimensions instead of 310; tests show the Fisher matrix and rate are unchanged.
- **ρ update.** One ρ per RSU is found by bisection on the objective, since the paper gives no update rule.

5.6 Conventional ISAC Benchmark

The benchmark of [2] minimises the trace of the CRB subject to a minimum rate per single-antenna user, solved by SDR and rank-one reconstruction. We implemented it as written in `baselines.py` and extended it to two RSUs and multiple targets. Our solver agrees with CVXPY within 0.25% wherever CVXPY reports an optimal solution.

5.7 Testing

The repository contains **189 automated tests**, all of which pass with the pinned library versions (re-run on 29 October 2026). Table 5.1 lists the main checks.

Table 5.1: Summary of automated tests.

Module	Main checks
Channel	Near-field vector tends to far field at large distance; Fresnel form matches exact phase
Scenario	Spacing and road bounds hold for every K and seed; sampler matches rejection sampling
Tracking	Jacobians match finite differences; noise statistics match Table III; filters exact without noise
CRB	Matches a Monte Carlo maximum-likelihood estimator; filters respect the posterior CRB
Beamforming	Lemma 2 rate never above true rate; subspace reduction exact; solver matches or beats CLARABEL; gradients match finite differences
Benchmark	Agrees with CVXPY within 0.25%

CHAPTER 6

Results and Discussion

All numbers in this chapter are from our own runs and are saved in `results/` of the repository. Values are the mean over five seeds with ± one standard deviation unless stated otherwise. Paper values were read from its figures by eye and are saved in `docs/paper_values/`.

6.1 Near-Field Region and Near/Far-Field Divergence

With Nt = 310 and fc = 50 GHz (λ = 6.0 mm), the Rayleigh distance $Nt^2\lambda/2$ is **288 m** ($2D^2/\lambda$ with the true aperture gives 286 m). The paper's "about 300 m" holds within 4%, and the whole road lies in the near field.

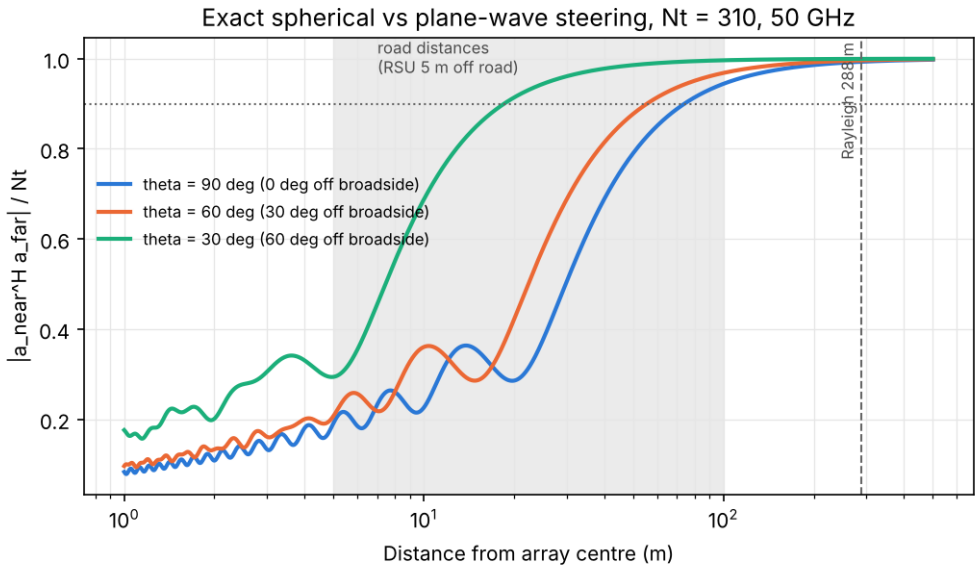


Figure 6.1: Correlation between the exact spherical and the plane-wave steering vectors against distance. Below 0.9 (dotted line) the far-field model is materially wrong.

- The plane-wave model falls below 0.9 correlation at **74 m** at broadside, **55 m** at 30° off broadside and **18.5 m** at 60° off broadside. Over random angles between 20° and 160° the mean divergence distance is 48.3 m (std 1.3 m).
- The Fresnel form of Eq. (5) matches the exact phase with correlation above 0.997 beyond 10 m at every angle; the nearest lane is 7.5 m away, so Eq. (5) is valid for this road.
- A far-field beam loses more than 0.9 dB at 14.1% of road points (std 1.7%) with the RSU at the road end, and at 33.8% (std 2.9%) with the RSU at mid-road. The importance of the near field therefore depends on the RSU position, which the paper does not state.

6.2 Road Scenario

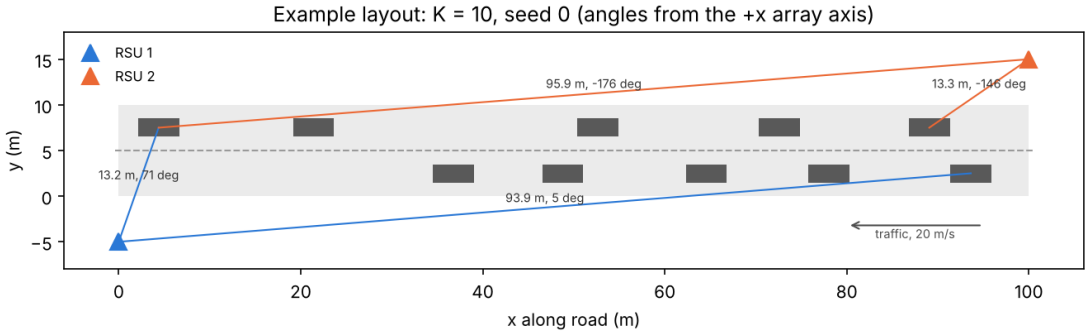


Figure 6.2: Example layout generated by the scenario module (K = 10 vehicles, seed 0).

At K = 15-20 the road is close to its capacity of 24 vehicles, so the layout is much more regular than a true Poisson process; the mean smallest gap is 3.72 m at K = 15 and 3.76 m at K = 20.

6.3 Tracking Performance

Reading the angle noise entries as rad² gives about 60° angle error for every filter, far from the paper's Fig. 3, so degree² was used. Taken literally, the β noise makes β grow about 2000 times in 2 s and the particle filter collapses (angle error 6-30°). The main results therefore use relative β noise; the as-written run is kept in the repository.

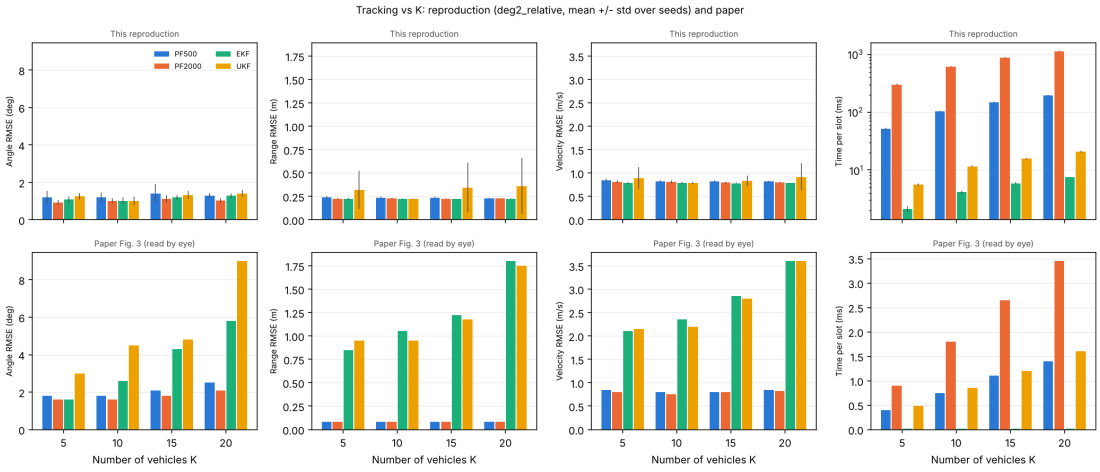


Figure 6.3: Tracking RMSE (angle, range, velocity) and computation time per slot against the number of vehicles: our results (top) and the paper's Fig. 3 read by eye (bottom).

Table 6.1: Tracking results compared with the paper.

Metric (K = 5 / K = 20)	Paper PF2000	Ours PF2000	Paper EKF	Ours EKF
Angle RMSE (deg)	1.6 / 2.1	0.93 / 1.03	1.6 / 5.8	1.09 / 1.28
Range RMSE (m)	0.08 / 0.08	0.226 / 0.227	0.85 / 1.8	0.226 / 0.226
Velocity RMSE (m/s)	0.80 / 0.82	0.81 / 0.80	2.1 / 3.6	0.79 / 0.79
Time per slot (ms)	0.9 / 3.45	300 / 1127	≈ 0 / ≈ 0	2.1 / 7.5

- The particle filter's velocity RMSE of about 0.8 m/s is **reproduced**.
- Angle and range RMSE are **not reproduced**: our angle error is lower (0.93-1.11°) and our range error higher (0.23 m against 0.08 m). Our range error sits at the measurement noise level.
- The paper's finding that EKF and UKF are much worse than the particle filter is **not reproduced**: ours are about as accurate and nearly flat in K. With a 310-element array the echo is strong, which keeps the linearised filters accurate; the paper does not explain what makes its EKF degrade with K.
- The ordering of computation time (EKF fastest, PF2000 slowest) is **reproduced**; absolute times are 100-1000 times larger because each measurement is a 310-element vector and our code is Python on one core.

6.4 Cramér-Rao Bound

A Monte Carlo maximum-likelihood estimator gives a ratio of mean squared error to CRB of 0.99 ± 0.12 for angle alone and 1.10 and 1.06 for joint angle and distance, which confirms that the corrected bound is computed correctly.

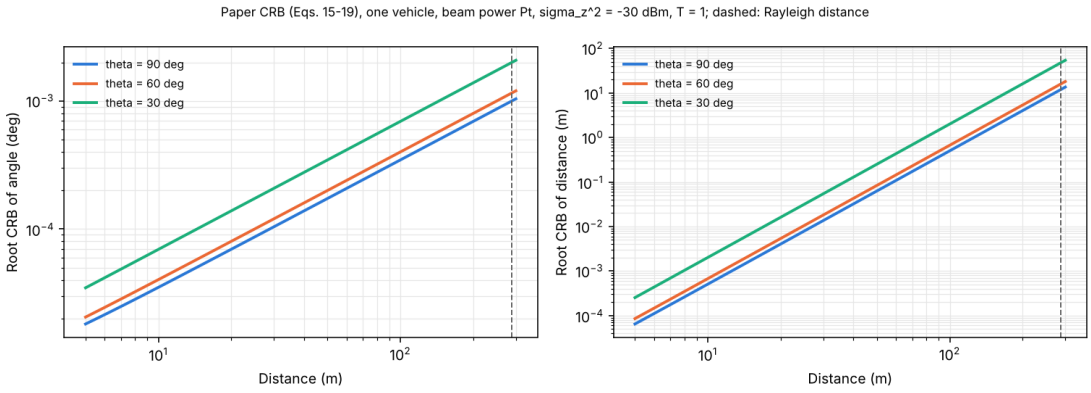


Figure 6.4: Root CRB of angle and distance against distance for one vehicle. Both grow in proportion to distance.

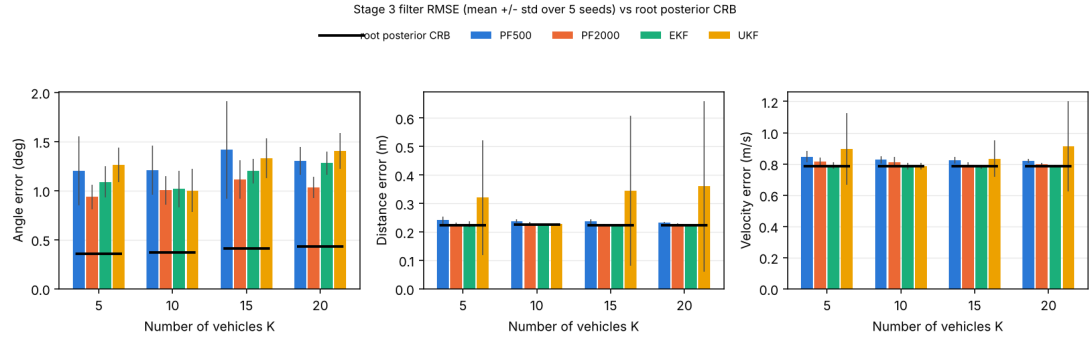


Figure 6.5: Filter RMSE against the posterior CRB.

Against the posterior CRB, the filters' angle RMSE is 1.8-3.6 times the bound, while distance and velocity lie on the bound (pooled ratio 0.996-2.0), as theory predicts for the linear Gaussian

part of the state. We also note that the paper uses −30 dBm for echo noise in the CRB but 0.04 in the tracking model, a difference of 46 dB.

6.5 Assignment and Beamforming

With Eq. (24) as written, a 1 m distance RCRB adds 1.2×10^8 cycles, which needs 8 GHz for one vehicle against a limit of 5.8 GHz. All 80 final assignments of every scheme violate this constraint, so problem (34) has no feasible point as written; the results below use L = |v3|.

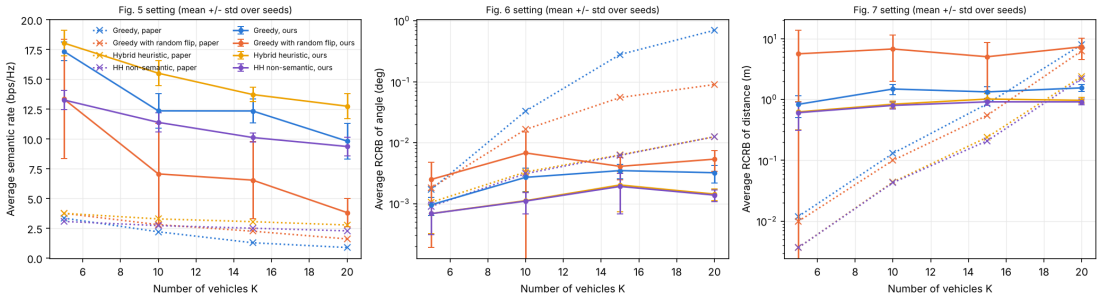


Figure 6.6: Semantic rate (paper's Fig. 5), angle RCRB (Fig. 6) and distance RCRB (Fig. 7) against the number of vehicles: ours (solid) and the paper (dotted).

Table 6.2: Rate and RCRB compared with the paper.

K	Scheme	Rate (bps/Hz)	Paper	Angle RCRB (deg)	Paper	Distance RCRB (m)	Paper
5	HH	18.0 ± 1.1	3.75	7×10^{-4}	1.0×10^{-3}	0.62	3.7×10^{-3}
5	Greedy	17.3 ± 0.8	3.35	1.0×10^{-3}	1.7×10^{-3}	0.83	1.2×10^{-2}
20	HH	12.7 ± 1.0	2.77	1.4×10^{-3}	1.25×10^{-2}	0.97	2.4
20	Greedy	9.8 ± 1.5	0.88	3.2×10^{-3}	0.7	1.55	8.0
20	Greedy + flip	3.8 ± 1.2	1.6	5.4×10^{-3}	9×10^{-2}	7.3	6.3

- The hybrid heuristic beats greedy assignment at every K on rate and both RCRBs, and the gap grows with K (4% at K = 5, 30% at K = 20): **reproduced**.
- The rate level is **not reproduced**: ours is 4-5 times the paper's, because the path loss of Eq. (6) has no wavelength term and our link SNR is about 50 dB.
- Angle RCRB is of the same order as the paper for HH (**partly reproduced**); distance RCRB is about 1 m and flat in K (**not reproduced**).
- Greedy with random flip is the worst scheme in our runs, whereas it beats greedy in the paper.
- ρ always settles at its lower bound 0.81 for HH, so the rate without semantics is exactly 1 − 0.81/1.1 = 26.5% lower (paper: 18%).

6.6 Computing Frequency and Digital Twin Power

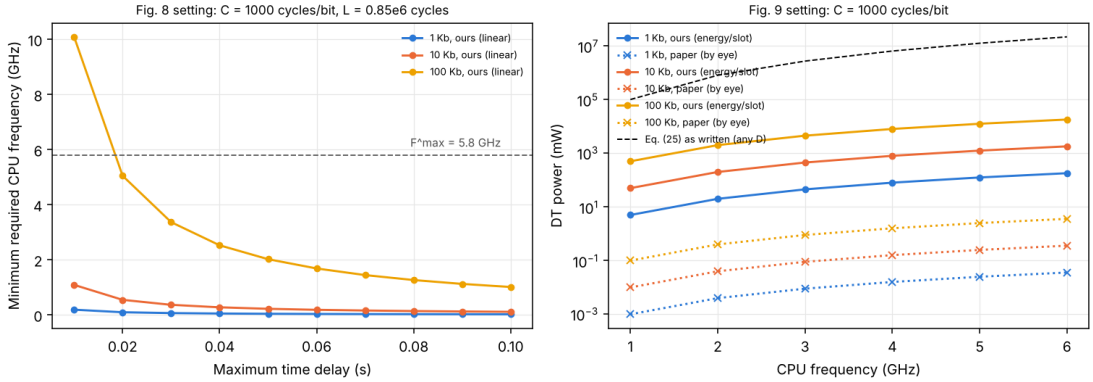


Figure 6.7: CPU frequency needed against allowed delay (paper's Fig. 8) and digital twin power against CPU frequency (paper's Fig. 9).

Fig. 8 is **reproduced** with C = 1000 cycles/bit: at 0.01 s we need 10.1, 1.09 and 0.19 GHz for 100, 10 and 1 Kb (paper about 10, 1 and 0.1 GHz). Fig. 9 is **partly reproduced**: the shape (power grows as f^2 and in proportion to D) matches, but our level is exactly 5000 times the paper's; the paper's level would need C = 0.2 cycles/bit, outside Table III's range.

6.7 The Headline Claim at 1, 10 and 100 Kb

Every vehicle's task was given the same size D. Schemes compared: HH; HH non-semantic; HH with one receive antenna (Nr = 1); and the ISAC benchmark of [2].

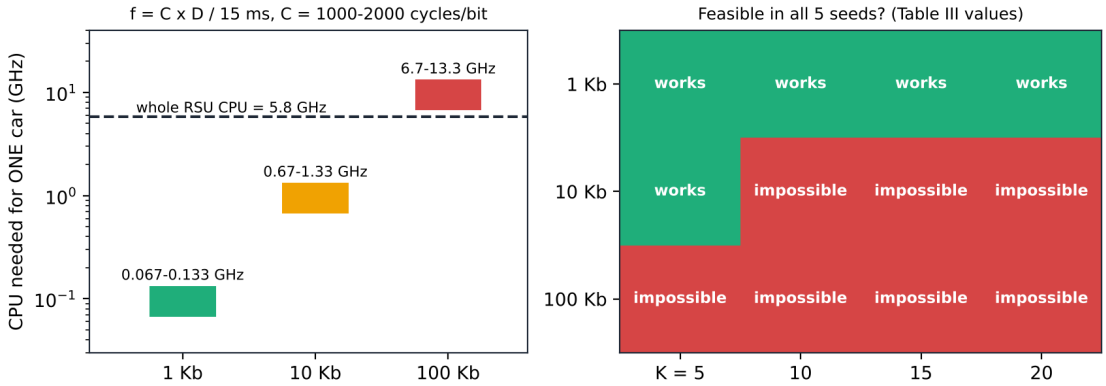


Figure 6.8: CPU needed by one vehicle within 15 ms, and feasibility of each case with Table III values.

Table 6.3: Feasibility at the three data sizes (identical for all five seeds).

Data size	K = 5	K = 10 to 20	Reason
1 Kb	Feasible	Feasible	-
10 Kb	Feasible	Infeasible	One vehicle needs 0.67-1.33 GHz of 5.8 GHz, so an RSU can serve only 4-8 vehicles
100 Kb	Infeasible	Infeasible	$CD/T^{max} = 1000 \times 10^5/0.015 = 6.7$ GHz for one vehicle, above 5.8 GHz

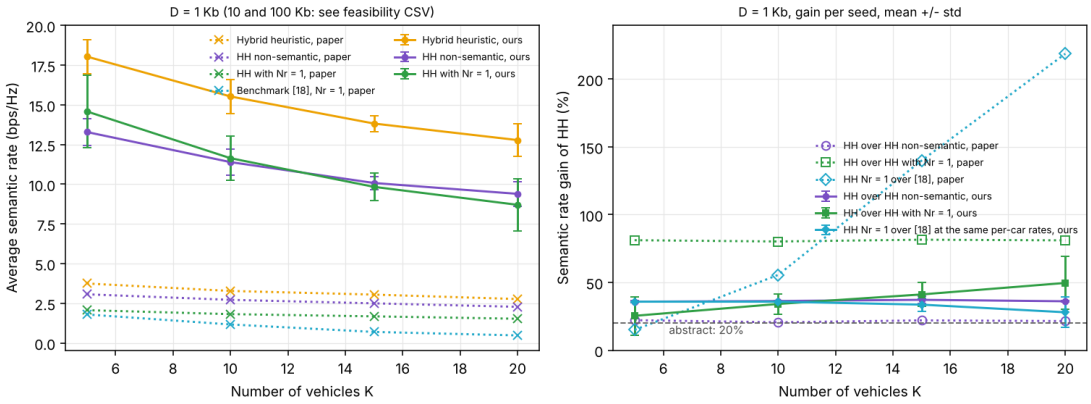


Figure 6.9: Semantic rates at 1 Kb (left) and rate gain of HH over each baseline (right); the dashed line marks the claimed 20%.

Table 6.4: Semantic rate gain at 1 Kb.

K	HH over HH non-semantic	Paper (Fig. 5)	HH over HH, Nr = 1	Paper (Fig. 5)
5	35.8 ± 0.6%	22%	25 ± 14%	81%
10	36.3 ± 0.3%	21%	34 ± 8%	80%
15	37.2 ± 2.4%	22%	41 ± 9%	82%
20	36.1 ± 0.6%	21%	50 ± 20%	81%

- Equal sensing accuracy holds against the non-semantic design (RCRB ratios 1.01-1.07 for angle, 1.03-1.06 for distance).
- The gain is **36%, not 20%**. It equals $\iota/\rho_{LB} - 1 = 1.1/0.81 - 1 = 35.8\%$, because ρ always reaches its lower bound; it is fixed by two Table III values, not by the optimisation.
- Against the benchmark [2]: HH with three receive antennas senses better than the benchmark at every rate target tried. Given HH (Nr = 1)'s own per-vehicle rates, the benchmark meets them with a 3-25% smaller CRB sum. HH's advantage at equal or better accuracy is therefore again only ι/ρ: 36% at K = 5 and 10, 28 ± 11% at K = 20. The paper shows 15% rising to 219%.
- At 10 Kb (K = 5 only) the gain over the benchmark at matched rates is 14 ± 7%, since the twin's CPU power leaves little budget for semantic extraction and ρ rises towards 1. At 100 Kb no feasible point exists.

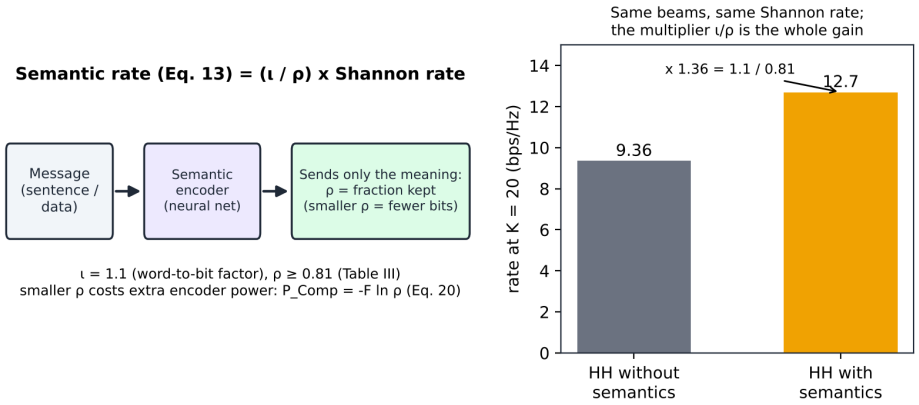


Figure 6.10: Semantic rate equals (ι/ρ) times the Shannon rate. At K = 20 the Shannon rates with and without semantics are equal (9.38 and 9.36 bps/Hz); the extra rate is only the multiplier.

Verdict. The headline claim is **not reproduced** as stated. Against both readings of "conventional ISAC", the rate gain at equal sensing accuracy comes entirely from the factor ι/ρ in Eq. (13). With Table III's values it is 36% at 1 Kb and 14% at 10 Kb, and 100 Kb is infeasible.

6.8 Measurements

The comparison framework document was not available, so we defined the measurement list ourselves: semantic and Shannon rate, angle and distance RCRB, CRB sum, CPU frequency, the three power terms, ρ, run times, and feasibility, for six schemes. Full tables are in Appendix B.

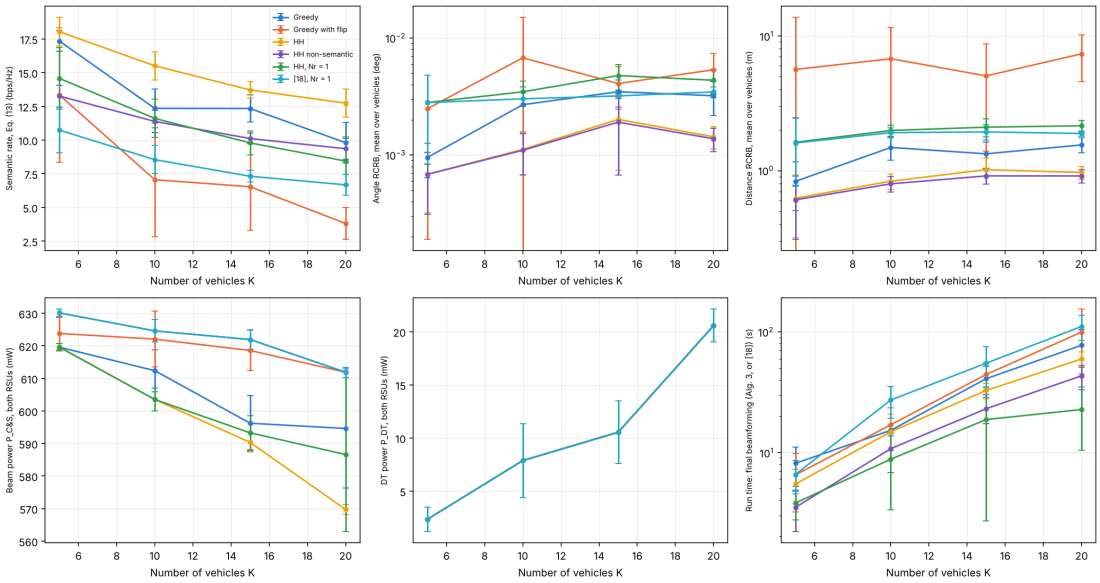


Figure 6.11: Measurements for six schemes against the number of vehicles.

- HH's Shannon rate equals HH non-semantic's (13.3 against 13.3 bps/Hz at K = 5; 9.38 against 9.36 at K = 20).

- At K = 20 (HH, both RSUs) the beams use 570 mW, semantic extraction 42.1 mW and the digital twin 20.6 mW; the busier RSU uses 2.28 GHz of CPU.
- On one core, HH assignment takes 3 s (K = 5) to 101 s (K = 20) and the final beamforming 5-60 s; the benchmark takes 7-111 s. The paper reports no optimisation run times.

6.9 Summary of Reproduction

Stage	What the paper says	What we got (5 seeds)	Verdict
1	Near-field zone ≈ 300 m	288 m	Reproduced
1	Near/far divergence range	74 m broadside, 18.5 m at 60° off	Our own result
3	PF angle RMSE 1.6-2.1°	0.93-1.11°	Not reproduced
3	PF range RMSE ≈ 0.08 m	0.23 m	Not reproduced
3	PF speed RMSE ≈ 0.8 m/s	0.80 m/s	Reproduced
3	EKF/UKF much worse than PF	about equal, flat in K	Not reproduced
3	PF(500) time < 2 ms at K = 20	194 ms (Python)	Not reproduced
4	CRB Eqs. 15-19 as printed	valid after 2 fixes	Partly
5	HH rate 3.75 / 2.77 bps/Hz	18.0 / 12.7 bps/Hz	Not reproduced
5	HH beats greedy, gap grows	4% → 30%	Reproduced
5	Greedy+flip beats greedy	it is the worst	Not reproduced
5	Angle RCRB HH 1e-3 / 1.3e-2°	7e-4 / 1.4e-3°	Partly
5	Distance RCRB 4 mm → 2.4 m	0.62 → 0.97 m	Not reproduced
5	Fig. 8 CPU, 100 Kb at 0.01 s ≈ 10 GHz	10.1 GHz	Reproduced
5	Fig. 9 DT power shape	same shape, 5000x level	Partly
6	Semantic gain 18-22% ("20%")	35.8-37.2%	Not reproduced
6	Semantic and plain HH sense equally	RCRB ratio 1.01-1.07	Reproduced

Figure 6.12: Summary of every claim checked against a number in the paper.

Of 16 checks against the paper, 5 are reproduced, 3 are partly reproduced and 8 are not reproduced. Two root causes explain most differences: the path loss of Eq. (6) has no wavelength term, which raises the SNR to about 50 dB; and ρ always reaches its lower bound of 0.81, which fixes the semantic gain at ι/ρ − 1 = 36%. Other differences arise from unstated details (RSU positions, noise and power units) and from constraints that cannot be met with Table III values. We also found two places where the paper's text and figures disagree (Fig. 6 at K = 20 for greedy, and Fig. 7 at K = 20 for HH).

6.10 Cost Comparison with Task 1

Task 1 results were not available to us, so only the RSU-side cost can be given. At K = 20 with HH, both RSUs together use 633 mW (beams 570 mW, semantic extraction 42.1 mW, digital twin CPU 20.6 mW), which is **12.7 mJ per 20 ms slot, or 0.63 mJ per vehicle per slot**, and 2.28 GHz of CPU on the busier RSU. These figures can be compared directly with the per-vehicle sensing and upload energy of Task 1. The beam power serves both communication and sensing.

CHAPTER 7

Conclusion and Future Scope

7.1 Conclusion

We reproduced the near-field ISCSC framework of Yang et al. in Python, stage by stage, directly from the paper's equations and parameter table, with every assumption recorded and every model tested. The broad picture of the paper holds: with 310 antennas at 50 GHz the whole road is in the near field (288 m); the far-field model fails within about 20-75 m depending on the angle; the particle filter tracks velocity to 0.8 m/s; and the hybrid heuristic outperforms greedy assignment by a margin that grows with the number of vehicles.

The headline claim does not reproduce as stated. At 1 Kb the semantic rate gain over conventional ISAC at equal sensing accuracy is 36%, not 20%, and it comes entirely from the multiplier ι/ρ rather than from the proposed beamforming or assignment, since the benchmark reaches equal rates with equal or better sensing accuracy. At 10 Kb the system works only for five vehicles, and at 100 Kb it is infeasible with the paper's own CPU and latency limits. Several printed equations needed correction, and some constraints cannot be met as written. These findings are as much a result of the reproduction as the matching curves.

7.2 Limitations

- Paper values were read from figures by eye, so differences below about 10% are not meaningful.
- Several parameters are our own assumptions (Table 4.4); other reasonable choices, particularly RSU positions and noise units, would change the absolute numbers.
- Stages 5-7 report one optimised slot per seed, because one HH run at K = 20 takes 2-3 minutes on one core.
- Run times are for Python on one core and are not comparable with the paper's, whose platform is not stated.
- The exponential workload curves of Fig. 8 were not reproduced, since their formula is not given.

7.3 Future Scope

- Obtain the works cited by the paper for the noise and power models to settle the unstated units.
- Repeat the study with a physical path loss that includes the wavelength.

- Run the optimisation over full 2 s trajectories with the tracked, rather than true, vehicle states.
- Complete the cost comparison with the device-side approach of Task 1.

References

[1] Y. Yang, Y. Ding, J. Wang, Z. Yang, C. Zhu, Z. Zhang, D. Niyato and M. Shikh-Bahaei, “Near-field integrated sensing, computing and semantic communication in digital twin-assisted vehicular networks,” *IEEE Trans. Veh. Technol.*, vol. 75, no. 9, 2026, doi: 10.1109/TVT.2026.3683229.

[2] Z. Wang, X. Mu and Y. Liu, “Near-field integrated sensing and communications,” *IEEE Commun. Lett.*, vol. 27, no. 8, 2023.

[3] P. Tichavský, C. H. Muravchik and A. Nehorai, “Posterior Cramér-Rao bounds for discrete-time nonlinear filtering,” *IEEE Trans. Signal Process.*, vol. 46, no. 5, pp. 1386-1396, 1998.

[4] Project source code and results, github.com/yagnangg333/nf-iscsc-reproduction.

[5] Y. Liu, Z. Wang, J. Xu, C. Ouyang, X. Mu and R. Schober, “Near-field communications: A tutorial review,” *IEEE Open J. Commun. Soc.*, vol. 4, 2023.

[6] N. J. Gordon, D. J. Salmond and A. F. M. Smith, “Novel approach to nonlinear/non-Gaussian Bayesian state estimation,” *IEE Proc. F*, vol. 140, no. 2, pp. 107-113, 1993.

[7] E. A. Wan and R. van der Merwe, “The unscented Kalman filter for nonlinear estimation,” in *Proc. IEEE Adaptive Systems for Signal Processing, Communications and Control Symp.*, 2000.

[8] Z.-Q. Luo, W.-K. Ma, A. M.-C. So, Y. Ye and S. Zhang, “Semidefinite relaxation of quadratic optimization problems,” *IEEE Signal Process. Mag.*, vol. 27, no. 3, pp. 20-34, 2010.

[9] S. Burer and R. D. C. Monteiro, “A nonlinear programming algorithm for solving semidefinite programs via low-rank factorization,” *Math. Program.*, vol. 95, pp. 329-357, 2003.

[10] S. Kirkpatrick, C. D. Gelatt and M. P. Vecchi, “Optimization by simulated annealing,” *Science*, vol. 220, no. 4598, pp. 671-680, 1983.

APPENDIX A

How to Run the Code

Python 3.11 is required. From the repository root:

```
python3.11 -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
pytest -q # all tests
```

Table A.1: Commands for each stage (times on 4 cores; set OMP_NUM_THREADS=1 for long runs).

Stage	Command	Time
1	`python experiments/stage1_nearfield_vs_farfield.py`	seconds
2	`python experiments/stage2_scenario.py`	seconds
3	`python experiments/stage3_tracking.py --angle-unit deg2 --beta-noise relative` `python experiments/stage3_plots.py deg2_relative`	≈ 15 min
4	`python experiments/stage4_crb.py`	≈ 3 min
5	`python experiments/stage5_allocation.py --workers 4 --tag main` `python experiments/stage5_plots.py --tag main`	≈ 1 h
5	`python experiments/stage5_computing.py`	seconds
6	`python experiments/stage6_semantic_gain.py --workers 4 --tag main`	≈ 40 min
6	`python experiments/stage6_benchmark18.py --workers 4 --tag main [--matched]` `python experiments/stage6_plots.py --tag main`	≈ 2 h
7	`python experiments/stage5_allocation.py --workers 4 --tag stage7` `python experiments/stage7_measurements.py --tag stage7`	≈ 2 h

APPENDIX B

Measurement Tables

Source: `results/stage7_measurements.md`. Mean ± std over five seeds, one optimised slot per seed, latency with L = |v3|. The benchmark [2] runs on the HH (Nr = 1) assignment with each vehicle's HH (Nr = 1) Shannon rate as its rate target.

Table B.1: Semantic rate, Eq. (13) (bps/Hz)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	17.3 ± 0.77	13.3 ± 5	18 ± 1.1	13.3 ± 0.81	14.6 ± 2.3	10.7 ± 1.7
10	12.3 ± 1.5	7.06 ± 4.2	15.5 ± 1.1	11.4 ± 0.81	11.6 ± 1.4	8.55 ± 1
15	12.3 ± 1	6.54 ± 3.3	13.7 ± 0.61	10.1 ± 0.39	9.79 ± 0.87	7.31 ± 0.42
20	9.81 ± 1.5	3.81 ± 1.2	12.7 ± 1	9.36 ± 0.78	8.44 ± 1.8	6.68 ± 0.77

Table B.2: Shannon rate (iota = rho = 1) (bps/Hz)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	12.8 ± 0.57	10.4 ± 3	13.3 ± 0.79	13.3 ± 0.81	10.7 ± 1.7	10.7 ± 1.7
10	9.87 ± 0.87	6.11 ± 3.3	11.4 ± 0.78	11.4 ± 0.81	8.55 ± 1	8.55 ± 1
15	9.39 ± 0.62	5.69 ± 2.4	10.1 ± 0.45	10.1 ± 0.39	7.31 ± 0.42	7.31 ± 0.42
20	8.1 ± 0.69	3.46 ± 1.1	9.38 ± 0.76	9.36 ± 0.78	6.67 ± 0.78	6.68 ± 0.77

Table B.3: Angle RCRB, mean over vehicles (deg)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	0.000955 ± 0.00031	0.00249 ± 0.0023	0.000685 ± 0.00037	0.000684 ± 0.00037	0.00281 ± 0.002	0.00281 ± 0.002
10	0.0027 ± 0.0011	0.00678 ± 0.0083	0.00112 ± 0.00044	0.0011 ± 0.00043	0.00347 ± 0.00084	0.00302 ± 0.00044
15	0.00348 ± 0.00091	0.00409 ± 0.0016	0.00202 ± 0.0013	0.00191 ± 0.0012	0.00478 ± 0.0012	0.0032 ± 0.00018
20	0.00321 ± 0.001	0.00536 ± 0.002	0.00143 ± 0.00031	0.00138 ± 0.00031	0.00436 ± 0.001	0.00347 ± 0.00036

Table B.4: Distance RCRB, mean over vehicles (m)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	0.833 ± 0.33	5.62 ± 8.1	0.625 ± 0.31	0.608 ± 0.3	1.62 ± 0.84	1.6 ± 0.84
10	1.49 ± 0.29	6.75 ± 4.8	0.835 ± 0.11	0.799 ± 0.1	1.99 ± 0.19	1.91 ± 0.16
15	1.33 ± 0.29	5.04 ± 3.6	1.02 ± 0.22	0.914 ± 0.12	2.1 ± 0.31	1.94 ± 0.26
20	1.55 ± 0.2	7.34 ± 2.8	0.969 ± 0.11	0.913 ± 0.1	2.15 ± 0.21	1.88 ± 0.083

Table B.5: Distance RCRB, median over vehicles (m)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	0.298 ± 0.32	0.672 ± 0.81	0.216 ± 0.2	0.215 ± 0.2	0.703 ± 0.56	0.697 ± 0.55
10	0.416 ± 0.21	1.96 ± 1.2	0.399 ± 0.24	0.392 ± 0.24	1.16 ± 0.58	1.16 ± 0.57
15	0.339 ± 0.12	0.953 ± 1	0.314 ± 0.09	0.325 ± 0.094	1.15 ± 0.27	1.31 ± 0.35
20	0.284 ± 0.032	1.56 ± 0.68	0.311 ± 0.051	0.303 ± 0.05	0.895 ± 0.11	0.831 ± 0.028

Table B.6: Sum of CRBs (eta in (34)) (m^2 + rad^2)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	10.5 ± 5.9	1.22e+03 ± 2.4e+03	5.42 ± 3.8	5.11 ± 3.6	34.8 ± 25	34 ± 24
10	84.5 ± 29	2.68e+03 ± 2.8e+03	16.2 ± 3.5	14.6 ± 3.1	83.3 ± 13	77.6 ± 12
15	103 ± 41	2.58e+03 ± 3.1e+03	49.3 ± 26	35.9 ± 8.4	167 ± 51	133 ± 37
20	211 ± 56	5.15e+03 ± 3.7e+03	57.4 ± 15	50.5 ± 13	247 ± 60	178 ± 20

Table B.7: CPU frequency used, busier RSU (GHz)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	0.619 ± 0.21	0.525 ± 0.1	0.551 ± 0.19	0.551 ± 0.19	0.56 ± 0.16	0.56 ± 0.16
10	1.13 ± 0.22	1.21 ± 0.25	1.11 ± 0.19	1.11 ± 0.19	1.11 ± 0.19	1.11 ± 0.19
15	1.58 ± 0.2	1.67 ± 0.32	1.5 ± 0.21	1.52 ± 0.26	1.52 ± 0.26	1.52 ± 0.26
20	2.35 ± 0.084	2.66 ± 0.26	2.28 ± 0.081	2.28 ± 0.081	2.3 ± 0.1	2.3 ± 0.1

Table B.8: Beam power $P_{C\&S}$, both RSUs (mW)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	620 ± 1.1	624 ± 5	620 ± 1.1	630 ± 1.1	620 ± 1.1	630 ± 1.1
10	612 ± 6.5	622 ± 8.5	604 ± 3.5	625 ± 3.5	604 ± 3.5	625 ± 3.5
15	596 ± 8.5	619 ± 6.2	590 ± 3	622 ± 3	593 ± 5.2	622 ± 3
20	595 ± 18	612 ± 1.5	570 ± 1.5	612 ± 1.5	587 ± 24	612 ± 1.5

Table B.9: Semantic extraction power P_{Comp}, both RSUs (mW)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	10.5 ± 0	6.32 ± 5.8	10.5 ± 0	0 ± 0	10.5 ± 0	0 ± 0
10	12.2 ± 5.2	2.53 ± 5.7	21.1 ± 0	0 ± 0	21.1 ± 0	0 ± 0
15	25.7 ± 8.1	3.37 ± 7.5	31.6 ± 0	0 ± 0	28.7 ± 6.6	0 ± 0
20	17.3 ± 18	0 ± 0	42.1 ± 0	0 ± 0	25.3 ± 23	0 ± 0

Table B.10: DT power P_{DT}, both RSUs (mW)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	2.35 ± 1.1	2.35 ± 1.1	2.35 ± 1.1	2.35 ± 1.1	2.35 ± 1.1	2.35 ± 1.1
10	7.88 ± 3.5	7.88 ± 3.5	7.88 ± 3.5	7.88 ± 3.5	7.88 ± 3.5	7.88 ± 3.5
15	10.5 ± 3	10.5 ± 3	10.5 ± 3	10.5 ± 3	10.5 ± 3	10.5 ± 3
20	20.6 ± 1.5	20.6 ± 1.5	20.6 ± 1.5	20.6 ± 1.5	20.6 ± 1.5	20.6 ± 1.5

Table B.11: Semantic ratio rho, mean (-)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	0.81 ± 0	0.886 ± 0.1	0.81 ± 0	1 ± 0	0.81 ± 0	1 ± 0
10	0.89 ± 0.047	0.977 ± 0.051	0.81 ± 0	1 ± 0	0.81 ± 0	1 ± 0
15	0.845 ± 0.049	0.98 ± 0.045	0.81 ± 0	1 ± 0	0.828 ± 0.04	1 ± 0
20	0.922 ± 0.08	1 ± 0	0.81 ± 0	1 ± 0	0.886 ± 0.1	1 ± 0

Table B.12: Run time: assignment (Alg. 2) (s)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	0 ± 0	0 ± 0	2.92 ± 0.9	3.11 ± 0.67	1.53 ± 0.39	0 ± 0
10	0 ± 0	0 ± 0	17.7 ± 2.8	18.3 ± 2.9	8.1 ± 3	0 ± 0
15	0 ± 0	0 ± 0	39.9 ± 9.1	40.9 ± 11	16.5 ± 3.4	0 ± 0
20	0 ± 0	0 ± 0	101 ± 13	115 ± 27	61.4 ± 17	0 ± 0

Table B.13: Run time: final beamforming (Alg. 3, or [2]) (s)

K	Greedy	Greedy with flip	HH	HH non-semantic	HH, Nr = 1	[2], Nr = 1
5	8.12 ± 2.9	6.5 ± 3.3	5.47 ± 1.7	3.49 ± 1.3	3.8 ± 1	6.56 ± 2
10	15.3 ± 1.5	17 ± 6.6	14.8 ± 6	10.7 ± 4	8.79 ± 5.4	27.3 ± 8.1
15	41 ± 11	44.4 ± 12	32.7 ± 4.7	23.1 ± 5.7	18.8 ± 16	55.1 ± 21
20	77.6 ± 27	99.8 ± 55	59.7 ± 8.6	43.3 ± 9.9	22.7 ± 12	111 ± 26

Table B.14: Feasibility at fixed data sizes (fraction of 5 seeds, Stage 6)

D (Kb)	K	HH	HH non-semantic	HH, Nr = 1
1	5	1.0	1.0	1.0
1	10	1.0	1.0	1.0
1	15	1.0	1.0	1.0
1	20	1.0	1.0	1.0
10	5	1.0	1.0	1.0
10	10	0.0	0.0	0.0
10	15	0.0	0.0	0.0
10	20	0.0	0.0	0.0
100	5	0.0	0.0	0.0
100	10	0.0	0.0	0.0
100	15	0.0	0.0	0.0
100	20	0.0	0.0	0.0